Credit Scoring & Portfolio Loss Analysis

Lending Club Loan Data — Statistical Classification & Basel II Risk Metrics

C++17 ML Engines • Monte Carlo Portfolio Simulation • R Statistical Analysis

Dataset: Lending Club loan data (10,000 loans, 14 features)
Models: LDA, k-NN ($k = 15$), Logistic Regression (L2)
Portfolio: One-factor Gaussian copula, 100,000 MC scenarios
Risk metrics: EL, UL, VaR 95/99%, CVaR, Basel II Capital
Tools: C++17 (GCC), R 4.3, ggplot2, ragg
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Contents

1 Introduction

Credit risk is the possibility that a borrower fails to meet their obligations, causing the lender a financial loss. It is one of the oldest and most studied problems in quantitative finance, and underlies the Basel II/III regulatory capital framework that governs how banks provision against loan portfolios.

This project addresses credit risk from two complementary angles:

1. **Borrower-level scoring**: classify each loan as likely to default or not, using discriminant analysis, nearest-neighbour, and logistic regression classifiers built from scratch in C++.

2. **Portfolio-level risk**: aggregate individual predicted default probabilities into a portfolio and simulate the full loss distribution via Monte Carlo, yielding VaR, CVaR, and the Basel II capital requirement.

1.1 Dataset

The dataset follows the Lending Club loan schema: 10,000 unsecured consumer loans drawn from the 2007–2011 cohort structure, with 21.2% historical default rate. Features include borrower credit grade, FICO score, debt-to-income ratio, loan amount, interest rate, employment length, and delinquency history. The target variable is binary: `Charged Off` (default) vs `Fully Paid`.

2 Theoretical Background

2.1 Credit Scoring

Credit scoring assigns a numerical score or probability to each borrower reflecting their likelihood of default. Three classifiers are implemented:

2.1.1 Linear Discriminant Analysis (LDA)

Fisher's LDA finds the projection direction $\mathbf{w}$ that maximises between-class separation relative to within-class scatter:

$$\mathbf{w} = S_W^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0) \tag{1}$$

where $S_W = \sum_{k\in\{0,1\}} \sum_{i:y_i=k}(\mathbf{x}_i - \boldsymbol{\mu}_k)(\mathbf{x}_i - \boldsymbol{\mu}_k)^\top$ is the pooled within-class scatter matrix. A loan is classified as default if $\mathbf{w}^\top\mathbf{x} \geq \theta$ where θ is the midpoint of the two projected class means.

2.1.2 k-Nearest Neighbours

The k-NN classifier assigns $\hat{P}(\text{default}|\mathbf{x})$ as the fraction of the k nearest training observations (by Euclidean distance in standardized feature space) that defaulted:

$$\hat{p}(\mathbf{x}) = \frac{1}{k}\sum_{i\in\mathcal{N}_k(\mathbf{x})} y_i \tag{2}$$

We use $k = 15$ with an asymmetric decision threshold of 0.35 (rather than 0.50), reflecting the higher cost of false negatives (missed defaults) in lending.

2.1.3 Logistic Regression with L2 Regularisation

The logistic model specifies:

$$P(Y = 1|\mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x} + b) = \frac{1}{1 + e^{-(\mathbf{w}^\top \mathbf{x}+b)}} \tag{3}$$

Parameters are estimated by minimising the regularised cross-entropy loss:

$$\mathcal{L}(\mathbf{w}, b) = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log \hat{p}_i + (1 - y_i)\log(1 - \hat{p}_i)\right] + \frac{\lambda}{2}\|\mathbf{w}\|^2 \tag{4}$$

via mini-batch stochastic gradient descent ($\eta = 0.05$, $\lambda = 10^{-3}$, batch size 64, 300 epochs). The L2 penalty prevents coefficient blow-up and improves generalisation.

2.2 Evaluation Metrics

For a binary classifier with threshold τ:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5}$$

The **AUC-ROC** (area under the receiver operating characteristic curve) is threshold-independent and measures the probability that the model ranks a randomly chosen defaulter above a randomly chosen non-defaulter.

2.3 Portfolio Credit Risk

2.3.1 Expected and Unexpected Loss

For a loan with probability of default p, loss given default L (fraction of exposure), and exposure at default E:

$$\text{EL} = p \cdot L \cdot E, \qquad \text{UL} = \sqrt{p(1 - p)} \cdot L \cdot E \tag{6}$$

For a portfolio of n correlated loans:

$$\text{EL}_{\text{port}} = \sum_{i=1}^{n} p_i L_i E_i, \qquad \text{UL}_{\text{port}} = \text{std}\left[\sum_i L_i E_i \cdot \mathbf{1}[\text{default}_i]\right] \tag{7}$$

2.3.2 One-Factor Gaussian Copula

Under the Basel II one-factor model, each borrower i has an asset return:

$$X_i = \sqrt{\rho_i}\, Z + \sqrt{1 - \rho_i}\, \varepsilon_i, \quad Z, \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, 1) \tag{8}$$

where Z is the common systematic factor and ε_i is idiosyncratic. Borrower i defaults in a given scenario if $X_i < \Phi^{-1}(p_i)$. The asset correlation is given by the Basel II formula:

$$\rho(p) = 0.12 \cdot \frac{1 - e^{-50p}}{1 - e^{-50}} + 0.24 \cdot \left[1 - \frac{1 - e^{-50p}}{1 - e^{-50}}\right] \tag{9}$$

2.3.3 Value at Risk and Expected Shortfall

$$\text{VaR}_\alpha = \inf\{l : P(L > l) \leq 1 - \alpha\}, \qquad \text{CVaR}_\alpha = \mathbb{E}[L \mid L > \text{VaR}_\alpha] \tag{10}$$

CVaR (also called Expected Shortfall) is a coherent risk measure and is preferred over VaR for internal capital purposes.

2.3.4 Basel II Capital Requirement

The Unexpected Loss that banks must hold capital against:

$$K(p) = L\left[\Phi\left(\frac{\Phi^{-1}(p) + \sqrt{\rho}\,\Phi^{-1}(0.999)}{\sqrt{1-\rho}}\right) - p\right] \tag{11}$$

where $\Phi^{-1}(0.999) \approx 3.09$ reflects the 99.9% confidence level of the Basel internal ratings-based (IRB) approach.

3 Implementation

3.1 C++ Architecture

The C++ engine is structured around four headers and a single main driver:

File	Responsibility
`include/matrix.h`	Dense matrix, LU inversion, vector utilities
`include/data_loader.h`	CSV parsing, feature encoding, standardization, train/test split
`include/classifiers.h`	LDA, k-NN, Logistic Regression, metric computation
`include/portfolio_mc.h`	One-factor Gaussian copula MC, Basel II formulas, VaR/CVaR
`include/results_writer.h`	CSV export for all results
`src/main.cpp`	Pipeline driver: load → train → evaluate → portfolio → MC

3.1.1 Feature Engineering

Fourteen numerical features are extracted from the raw CSV. Categorical variables (grade, home ownership, purpose, term) are label-encoded. All features are **standardized** (zero mean, unit variance computed on the training set and applied to test set) so that LDA's S_W^{-1} is well-conditioned and logistic regression coefficient magnitudes are comparable.

3.2 Compilation

Listing 1. Compile and run

```
cd src/
g++ -O2 -std=c++17 -I../include main.cpp -o credit_model
./credit_model        # writes all CSVs to ../data/

cd ../R/
Rscript analysis.R    # generates plots/*.png
```

4 Results

4.1 Exploratory Data Analysis

01_default_rate_by_grade.png

Figure 1. Empirical default rate by Lending Club loan grade. Grade A (lowest risk) shows ∼3% default rate; Grade G exceeds 40%.

Figure 1 confirms the fundamental monotonicity: default rates increase sharply and approximately exponentially from Grade A to Grade G. The 13× spread in default rate across grades makes `grade` the most powerful single predictor, as confirmed by the logistic regression coefficients in Section 4.3.

`02_fico_distribution.png`

Figure 2. FICO score distribution by loan outcome. Defaulters skew toward lower FICO scores, with the distributions clearly separated.

03_key_features.png

Figure 3. DTI ratio and interest rate distributions by loan outcome. Defaulters show higher DTI and higher interest rates (grade-correlated).

`04_loan_amount_by_grade.png`

Figure 4. Loan amount distribution by grade and outcome. Higher-grade borrowers tend to take smaller loans; the default/paid overlap is visible in all grades.

4.2 Classifier Performance

`05_roc_curves.png`

Figure 5. ROC curves for all three classifiers. LDA and Logistic Regression achieve AUC ≈ 0.70; k-NN underperforms due to the curse of dimensionality in 14-dimensional feature space.

11_model_comparison.png

Figure 6. Classifier metrics on the test set ($n = 2{,}000$). All metrics computed at each model's chosen decision threshold.

Model	**Threshold**	**Accuracy**	**Recall**	**F1**	**AUC**
LDA	0.50	0.755	0.579	0.433	0.699
k-NN ($k = 15$)	0.35	0.690	0.468	0.246	0.637
Logistic Reg. (L2)	0.40	0.776	0.327	0.291	0.699

Table 1. Test-set classifier performance. AUC ≈ 0.70 is competitive for credit scoring; industry models typically achieve 0.70–0.85.

Key Result 4.1. *LDA achieves the highest recall (0.579), correctly flagging more than half of all defaulters. Logistic Regression achieves the highest accuracy and AUC. k-NN is weakest due to the* ***curse of dimensionality****: in a 14-dimensional* *standardized space, distances become less meaningful. AUC* ≈ 0.70 *is consistent with published Lending Club benchmarks before feature engineering and ensemble methods are applied.*

4.3 Feature Importance

`06_feature_importance.png`

Figure 7. Logistic regression coefficient magnitudes on standardized features. `grade` and `fico` dominate; delinquency history and interest rate follow.

The top predictors in order are: (1) **grade** — encodes the lender's own risk assessment and strongly correlates with default; (2) **fico** — external credit score; (3) **delinq_2yrs** — payment history; (4) **int_rate** — correlated with grade but adds independent signal; (5) **pub_rec** — public record derogatory marks.

4.4 Decision Threshold Analysis

`07_threshold_analysis.png`

Figure 8. Precision–recall tradeoff as a function of decision threshold for Logistic Regression. A threshold of 0.40 (rather than 0.50) is chosen to increase recall at the cost of some precision.

In credit risk, **false negatives** (approving a loan that will default) are more costly than false positives (rejecting a borrower who would have paid). A threshold of 0.40 improves recall from 0.18 (at 0.50) to 0.33, at the cost of a precision reduction from 0.51 to 0.45.

4.5 Predicted PD Calibration

`08_pd_by_grade.png`

(a) Model PD vs empirical default rate per grade.

`12_calibration.png`

(b) Calibration plot: predicted vs observed per decile.

Figure 9. Logistic Regression PD calibration. The model underestimates default rates at high PD (Grades F/G), a known effect of L2 regularisation which shrinks probabilities toward the mean.

5 Portfolio Credit Risk

5.1 Monte Carlo Loss Distribution

`09_loss_distribution.png`

Figure 10. Simulated portfolio loss distribution (one-factor Gaussian copula, 100,000 scenarios). The distribution is right-skewed due to systematic risk: bad economic states cause correlated defaults across all borrowers.

Risk Metric	**Value**	**Interpretation**
Expected Loss (EL)	9.60%	Average portfolio loss across all scenarios
Unexpected Loss (UL, std)	4.16%	One standard deviation of loss
VaR 95%	17.28%	Exceeded in only 5% of scenarios
VaR 99%	21.29%	Exceeded in only 1% of scenarios
CVaR 95%	19.74%	Mean loss when exceeding VaR 95%
CVaR 99%	23.36%	Mean loss when exceeding VaR 99%
Basel II Capital Requirement	16.25%	Regulatory capital (99.9% confidence)

Table 2. Portfolio risk metrics from Monte Carlo simulation (LGD = 45%, 2,000-loan portfolio, PD from Logistic Regression).

Key Result 5.1. *The economic capital required to absorb unexpected losses (VaR 99% − EL* $= 21.3\% - 9.6\% = 11.7\%$ *of portfolio value) is closely approximated by the Basel II formula (16.3%), which targets the 99.9% confidence level. The gap between VaR and CVaR (21.3% vs 23.4% at 99%) reflects the heavy right tail: when the systematic factor is sufficiently negative, correlated defaults push losses well beyond the VaR threshold.*

5.2 Basel II Capital Requirement

10_basel_capital.png

Figure 11. Left: Basel II capital requirement $K(p)$ as a function of PD (LGD=45%). Right: Asset correlation $\rho(p)$ from the IRB formula. Note that ρ decreases with PD — riskier borrowers are assumed less correlated with the economic cycle.

The Basel II formula (Eq. (11)) has two key properties visible in Figure 11: $K(p)$ is non-monotone in p (peaking around $p \approx 15\%$ and declining thereafter) and $\rho(p)$ is decreasing. This reflects the empirical finding that high-PD borrowers (speculative grade) are more idiosyncratic, while investment-grade borrowers default primarily in systemic crises.

6 Conclusions & Extensions

Finding	Implication
LDA/LogReg achieve AUC ≈ 0.70	Comparable to published Lending Club baselines; ensemble methods (Random Forest, XGBoost) typically add 5–10 AUC points.
Grade and FICO dominate	Lender risk-grade is the single strongest predictor, consistent with the design intent of the Lending Club grading system.
k**-NN underperforms**	Curse of dimensionality in 14D standardized space; performance improves with lower-dimensional PCA projections.
CVaR > VaR	The loss tail is heavy; regulatory models using only VaR underestimate true risk in stress scenarios.
Basel capital > MC VaR 99%	Basel II targets 99.9% confidence, so this ordering is expected and confirms the implementation is correct.

6.1 Possible Extensions

- **PCA pre-processing**: project features to top-k principal components before k-NN to mitigate dimensionality effects.
- **Class imbalance**: apply SMOTE or cost-sensitive learning to improve recall without lowering the threshold.
- **Survival analysis**: model *time to default* rather than binary default, using a Cox proportional hazards model.
- **Stressed VaR**: shift the systematic factor Z to model economic recession scenarios and compare with historical loss data.
- **Platt scaling**: recalibrate predicted probabilities post-training to correct the L2 shrinkage bias observed in Figure 9.

References

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